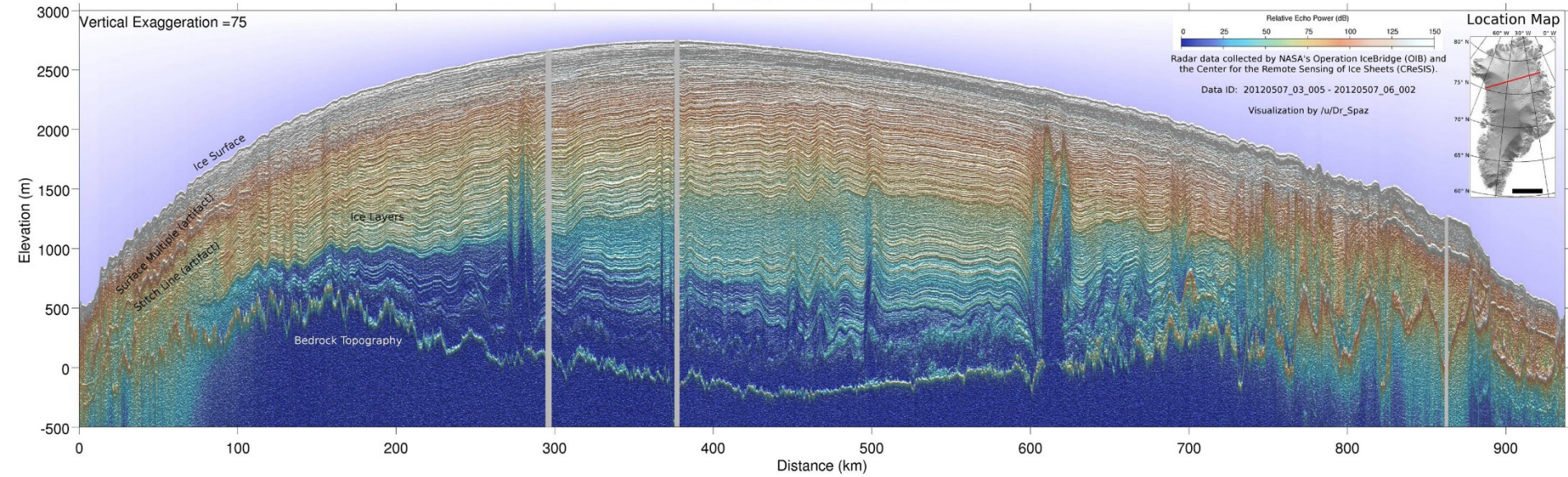


Cross-Section of the Greenland Ice Sheet



Lecture 6:

Ice sheet flow

Our ice-sheet model:

1. Mass conservation
2. Stress balance

$$\frac{\partial H}{\partial t} + \frac{\partial q}{\partial x} = \dot{b}_i$$

$$q = \frac{2A}{n+2} (\rho g \alpha)^n H^{n+2}$$

$$\dot{\epsilon} = \begin{bmatrix} \dot{\epsilon}_{xx} & \dot{\epsilon}_{xy} & \dot{\epsilon}_{xz} \\ \dot{\epsilon}_{yx} & \dot{\epsilon}_{yy} & \dot{\epsilon}_{yz} \\ \dot{\epsilon}_{zx} & \dot{\epsilon}_{zy} & \dot{\epsilon}_{zz} \end{bmatrix}$$

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix}$$

$$\dot{\epsilon}_{ij} = \frac{1}{2} \left[\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right]$$

$$\boldsymbol{\sigma} = \boldsymbol{\tau} + \boldsymbol{\sigma}_m$$

$$\dot{\epsilon}_{ij} = A \tau_E^{n-1} \tau_{ij} \quad (i, j = x, y, z)$$

$$\tau_E^2 = \frac{1}{2} [\tau_{xx}^2 + \tau_{yy}^2 + \tau_{zz}^2] + \tau_{xz}^2 + \tau_{xy}^2 + \tau_{yz}^2$$

$$\frac{\partial H}{\partial t} + \frac{\partial q}{\partial x} = b_i$$

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} = 0$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = \rho g$$

Recipe for deriving our shallow ice-approximation model

1. Derive a mass conservation equation
2. Stress balance equations $\rightarrow$ two dimensions
3. Express in terms of the ‘deviatoric stress’
4. Make (part of) the ‘shallow ice approximation’
5. Rearrange/combine/integrate until we have an equation for τ_{zx}
6. Bring in Glen’s flow law and definition of strain rate
7. Make the other part of the ‘shallow ice approximation’
8. Combine Glen’s flow law with our equation for τ_{zx}
9. Manipulate the result until we have an expressions for speed and for ice flux

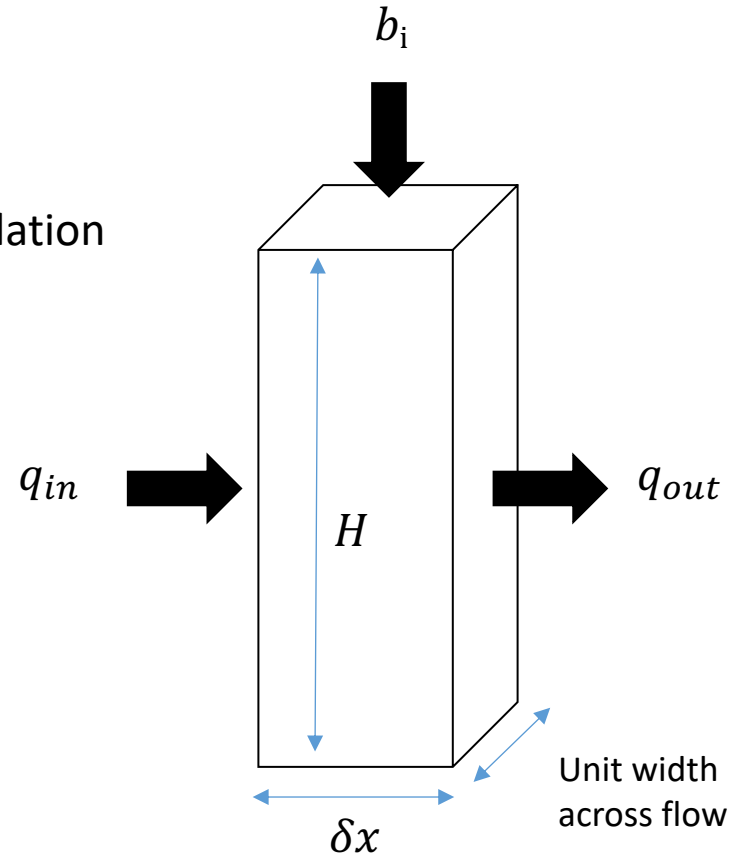
Mass conservation in a column of ice

ice thickness

volume flux
per unit width

$$\frac{\partial H}{\partial t} + \frac{\partial q}{\partial x} = \dot{b}_i$$

net accumulation



Mass conservation in a column of ice

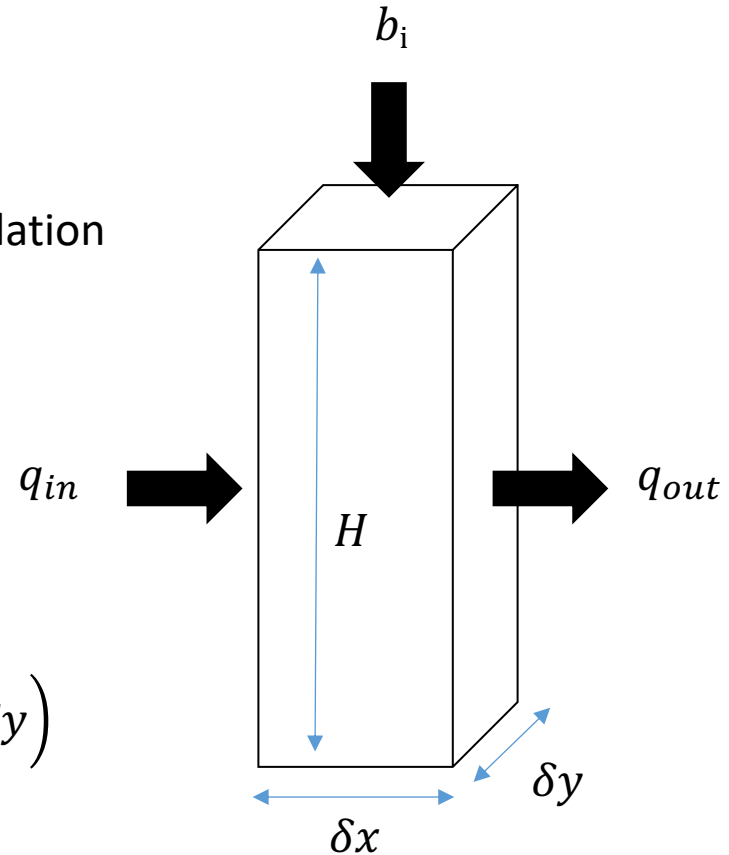
ice thickness

volume flux
per unit width

$$\frac{\partial H}{\partial t} + \frac{\partial q}{\partial x} = \dot{b}_i$$

net accumulation

$$\begin{aligned}\frac{\partial H}{\partial t} \delta x \delta y &= b_i \delta x \delta y + \delta y q_{in} - \delta y q_{out} \\ \frac{\partial H}{\partial t} \delta x \delta y &= b_i \delta x \delta y + \delta y q - \left(\delta y q + \frac{\partial q}{\partial x} \delta x \delta y \right) \\ \frac{\partial H}{\partial t} &= b_i - \frac{\partial q}{\partial x}\end{aligned}$$



Detailed derivation online here: https://Ideo-glaciology.github.io/glaciology-intro-book/sections/ice_flow/depth_integrated_mass_balance.html

A hierarchy of ice sheet models based on which terms you neglect

- Shallow Ice Approximation
- Shallow Shelf Approximation
- ‘Hybrid’
- ‘Higher-order’
- ‘Full-stokes’

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} = 0$$

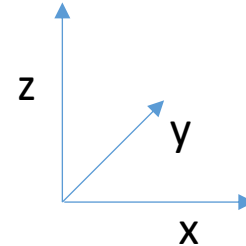
$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = \rho g$$

Stress-balance: Stokes equations

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} = 0$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = \rho g$$

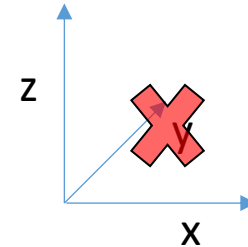


Stress-balance: 2D Stokes equations

$$\frac{\partial \sigma_{xx}}{\partial x} + \cancel{\frac{\partial \tau_{yx}}{\partial y}} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

$$\cancel{\frac{\partial \tau_{xy}}{\partial x}} + \cancel{\frac{\partial \sigma_{yy}}{\partial y}} + \cancel{\frac{\partial \tau_{zy}}{\partial z}} = 0$$

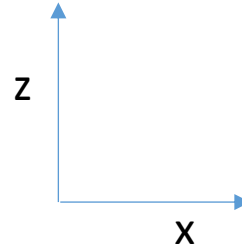
$$\frac{\partial \tau_{xz}}{\partial x} + \cancel{\frac{\partial \tau_{yz}}{\partial y}} + \frac{\partial \sigma_{zz}}{\partial z} = \rho g$$



Stress-balance: 2D Stokes equations

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

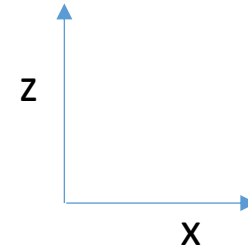
$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \sigma_{zz}}{\partial z} = \rho g$$



Stress-balance: 2D Stokes equations

$$\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \sigma_m}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{zz}}{\partial z} + \frac{\partial \sigma_m}{\partial z} = \rho g$$

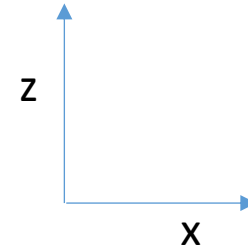


Deviatoric stress

$$\boldsymbol{\sigma} = \boldsymbol{\tau} + \sigma_m$$

Stress-balance: Shallow Ice Approximation

One part of the shallow ice approximation.



$$\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \sigma_m}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

Deviatoric stress

$$\boldsymbol{\sigma} = \boldsymbol{\tau} + \sigma_m$$

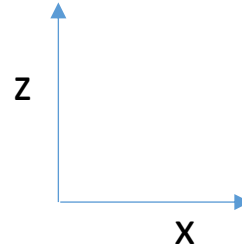
$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{zz}}{\partial z} + \frac{\partial \sigma_m}{\partial z} = \rho g$$

- 'shallow' refers to the idea that the aspect ratio is very large --> ice sheets are much wider than they are thick.
- In our model this leads to being able to neglect how some of the smaller stresses vary horizontally.
- Additionally we neglect the vertical variation of the deviatoric vertical normal stress.

Stress-balance: Shallow Ice Approximation

$$\frac{\partial \sigma_m}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

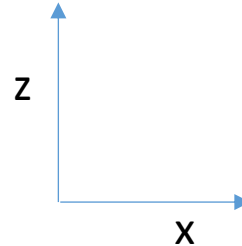
$$\frac{\partial \sigma_m}{\partial z} = \rho g$$



Deviatoric stress

$$\boldsymbol{\sigma} = \boldsymbol{\tau} + \sigma_m$$

Stress-balance: Shallow Ice Approximation



$$\frac{\partial \sigma_m}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

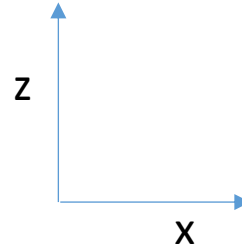
Integrate vertically

$$\sigma_m(z) = -\rho g(H - z)$$

Ice thickness

Using the
boundary
condition: $\sigma_m(H) = 0$

Stress-balance: Shallow Ice Approximation

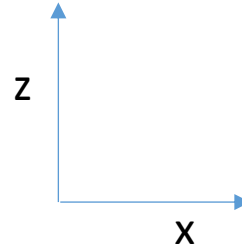


$$\frac{\partial \sigma_m}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

Differentiate horizontally

$$\frac{\partial \sigma_m}{\partial x} = -\rho g \frac{\partial H}{\partial x}$$

Stress-balance: Shallow Ice Approximation



$$\frac{\partial \sigma_m}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

Combine these two

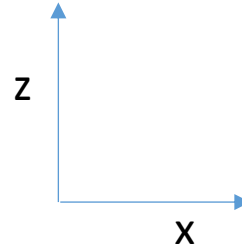


$$\frac{\partial \sigma_m}{\partial x} = -\rho g \frac{\partial H}{\partial x} \quad \Rightarrow \quad \frac{\partial \tau_{zx}}{\partial z} = \rho g \frac{\partial H}{\partial x}$$

Stress-balance: Shallow Ice Approximation

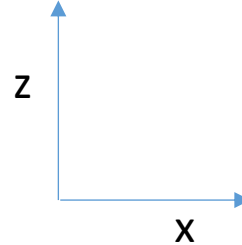
Vertical variation in vertical shear stress

$$\frac{\partial \tau_{zx}}{\partial z} = \rho g \frac{\partial H}{\partial x}$$



Stress-balance: Shallow Ice Approximation

Vertical variation in vertical shear stress



$$\frac{\partial \tau_{zx}}{\partial z} = \rho g \frac{\partial H}{\partial x}$$

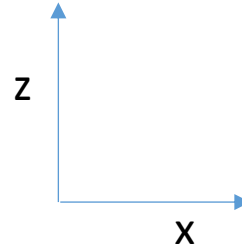
Integrate vertically

$$\tau_{zx} = -\rho g \frac{\partial H}{\partial x} (H - z)$$

Using the
boundary
condition: $\tau_{zx}(H) = 0$

Stress-balance: Shallow Ice Approximation

Vertical variation in vertical shear stress



$$\frac{\partial \tau_{zx}}{\partial z} = \rho g \frac{\partial H}{\partial x}$$

Re-arrange slightly to get:

$$\tau_{zx} = -\rho g \frac{\partial H}{\partial x} (H - z)$$

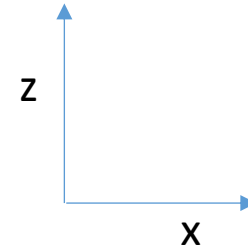
$$\tau_{zx} = \rho g \alpha H \left(1 - \frac{z}{H}\right)$$

where

$$\alpha = -\frac{\partial H}{\partial x}$$

Stress-balance: Shallow Ice Approximation

$$\tau_{zx} = \rho g \alpha H \left(1 - \frac{z}{H}\right)$$

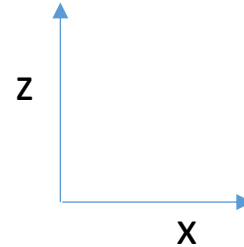


This expression evaluated at $z=H$ is called the driving stress

$$\tau_d = \rho g \alpha H$$

Stress-balance: Shallow Ice Approximation

$$\tau_{zx} = \rho g \alpha H \left(1 - \frac{z}{H}\right)$$



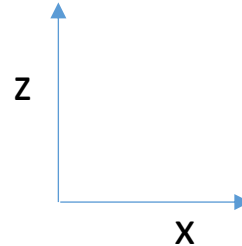
Glen's flow law

$$\dot{\epsilon}_{zx} = A \tau_{zx}^n$$

$$\tau_E = \tau_{xz}$$

Stress-balance: Shallow Ice Approximation

$$\tau_{zx} = \rho g \alpha H \left(1 - \frac{z}{H}\right)$$



Glen's flow law

$$\dot{\epsilon}_{zx} = A \tau_{zx}^n$$

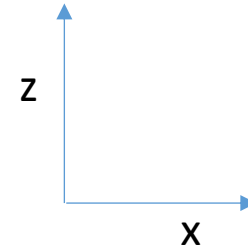
$$\tau_E = \tau_{xz}$$

Strain rate

$$\dot{\epsilon}_{zx} = \frac{1}{2} \left[\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right]$$

Stress-balance: Shallow Ice Approximation

$$\tau_{zx} = \rho g \alpha H \left(1 - \frac{z}{H}\right)$$



Glen's flow law

$$\dot{\epsilon}_{zx} = A \tau_{zx}^n$$

$$\tau_E = \tau_{xz}$$

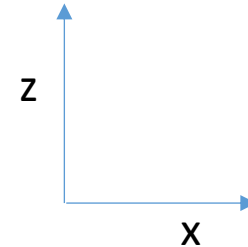
Strain rate

$$\dot{\epsilon}_{zx} = \frac{1}{2} \left[\cancel{\frac{\partial w}{\partial x}} + \frac{\partial u}{\partial z} \right]$$

$$\dot{\epsilon}_{zx} = \frac{1}{2} \frac{\partial u}{\partial z}$$

Stress-balance: Shallow Ice Approximation

$$\frac{\partial u}{\partial z} = 2A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n$$



Glen's flow law

$$\epsilon_{zx}^{\dot{}} = A \tau_{zx}^n$$

$$\tau_E = \tau_{xz}$$

Combine
these

Strain rate

$$\epsilon_{zx}^{\dot{}} = \frac{1}{2} \left[\cancel{\frac{\partial w}{\partial x}} + \frac{\partial u}{\partial z} \right]$$

$$\epsilon_{zx}^{\dot{}} = \frac{1}{2} \frac{\partial u}{\partial z}$$

Stress-balance: Shallow Ice Approximation

$$\frac{\partial u}{\partial z} = 2A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n$$
$$F = 1 - \frac{z}{H}$$

Integrate by substitution

$$u(z) = 2A(\rho g \alpha H)^n \left(\frac{H}{n+1}\right) \left(1 - \left(1 - \frac{z}{H}\right)^{n+1}\right)$$

Stress-balance: Shallow Ice Approximation

$$\frac{\partial u}{\partial z} = 2A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n \quad F = 1 - \frac{z}{H}$$

Integrate by substitution

$$u(z) = 2A(\rho g \alpha H)^n \left(\frac{H}{n+1}\right) \left(1 - \left(1 - \frac{z}{H}\right)^{n+1}\right)$$

Integrate one more time to get depth-integrated volume-flux per unit width:

$$q = \frac{2A}{n+2} (\rho g \alpha)^n H^{n+2}$$

- 2-min break

The complete model:

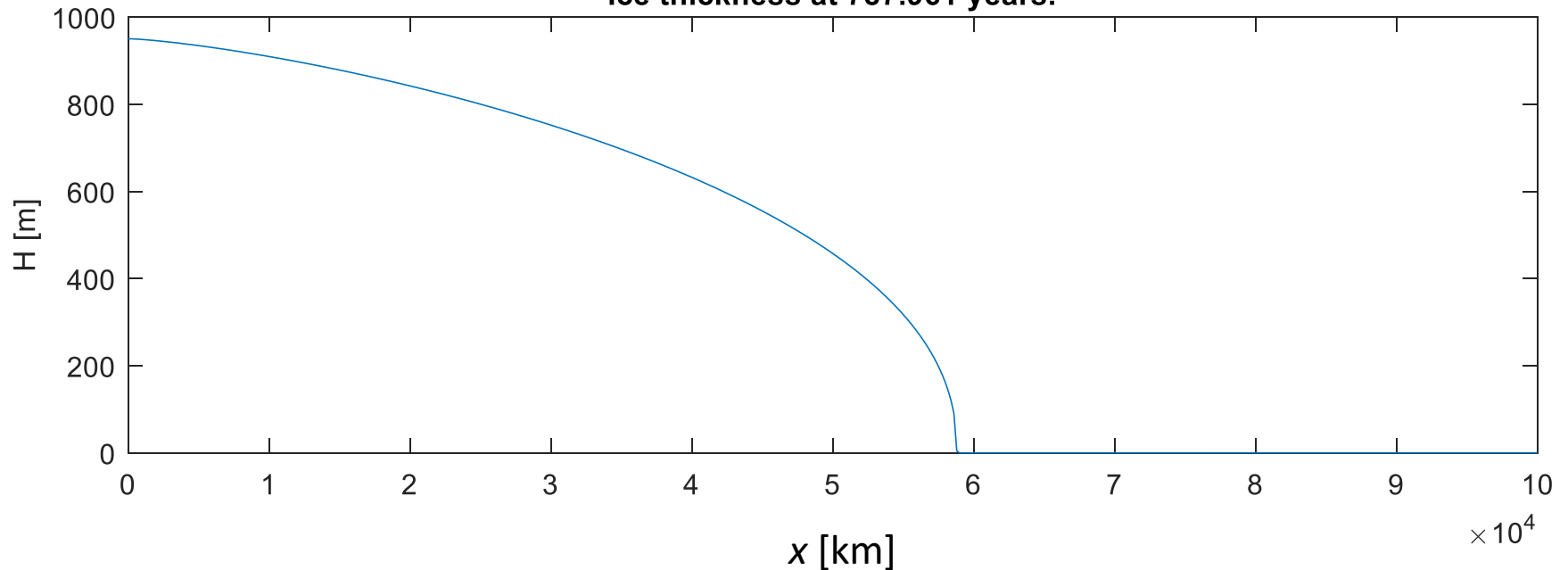
mass conservation:

$$\frac{\partial H}{\partial t} + \frac{\partial q}{\partial x} = \dot{b}_i$$

Depth-integrated volume-flux per unit width:

$$q = \frac{2A}{n+2} (\rho g \alpha)^n H^{n+2}$$

Ice thickness at 787.961 years.



Flux strongly depends on H . Why?

$$q = \frac{2A}{n+2} (\rho g \alpha)^n H^{n+2}$$

$q \propto H^{n+2}$ With $n = 3$ that's H^5 !!!

n comes from the nonlinear rheology and the driving stress being proportional to H

+1 comes from the fact that thicker ice flows faster

+1 comes from the fact that thicker ice moves more ice for a given flow speed

Summary

- Driving stress drives ice flow. It is proportional to ice thickness and surface slope.
- Ice sheet models use depth-integrated mass conservation to evolve the surface up and down.
 - a balance between flux coming in, flux going out and net specific accumulation.
- We derived a 'simple' equations for the flux
- Together the mass conservation and flux equations make up our ice sheet model. s

For Tuesday

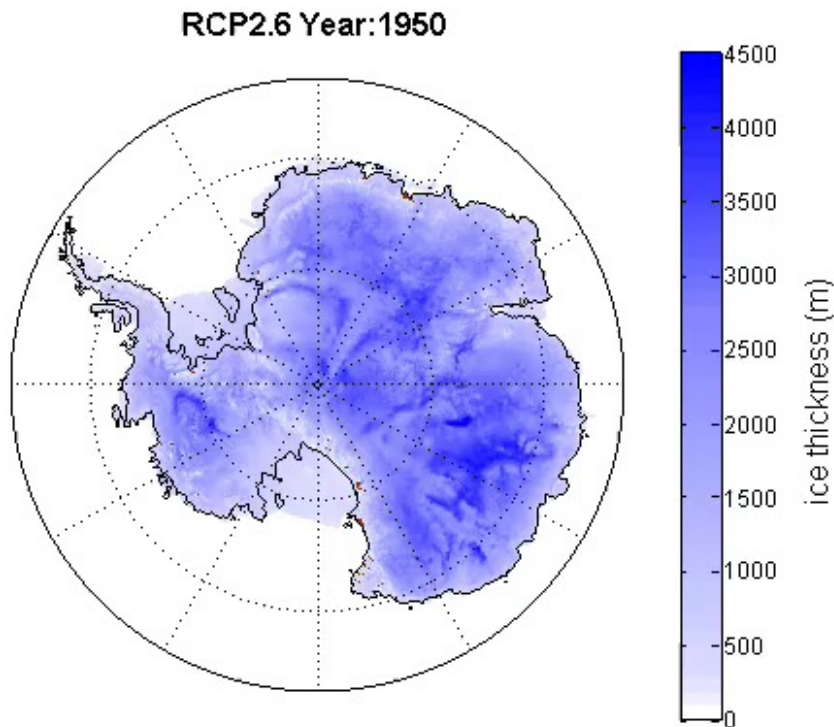
(1) go through these derivations and make sure you understand each step (you don't have to memorize anything, just make sure you understand each step).

(2) revise the “finite-difference method” of solving differential equations. There are many online resources: e.g.

https://www.ljll.math.upmc.fr/frey/cours/UdC/ma691/ma691_ch6.pdf

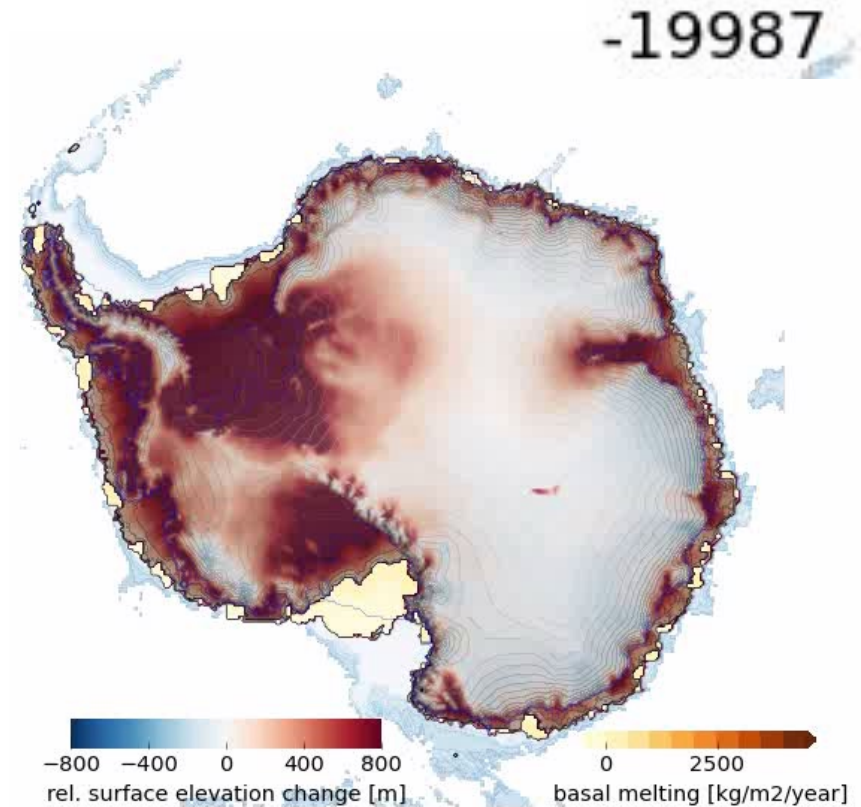
Two state-of-the-art hybrid models

Take into account GIA, ocean/atmosphere forcing, calving, surface mass balance, spatially-variable bed conditions.



DeConto and Pollard (2013)

Penn State University model



Kingslake et al. (*in prep.*)

Parallel Ice Sheet model (PISM)

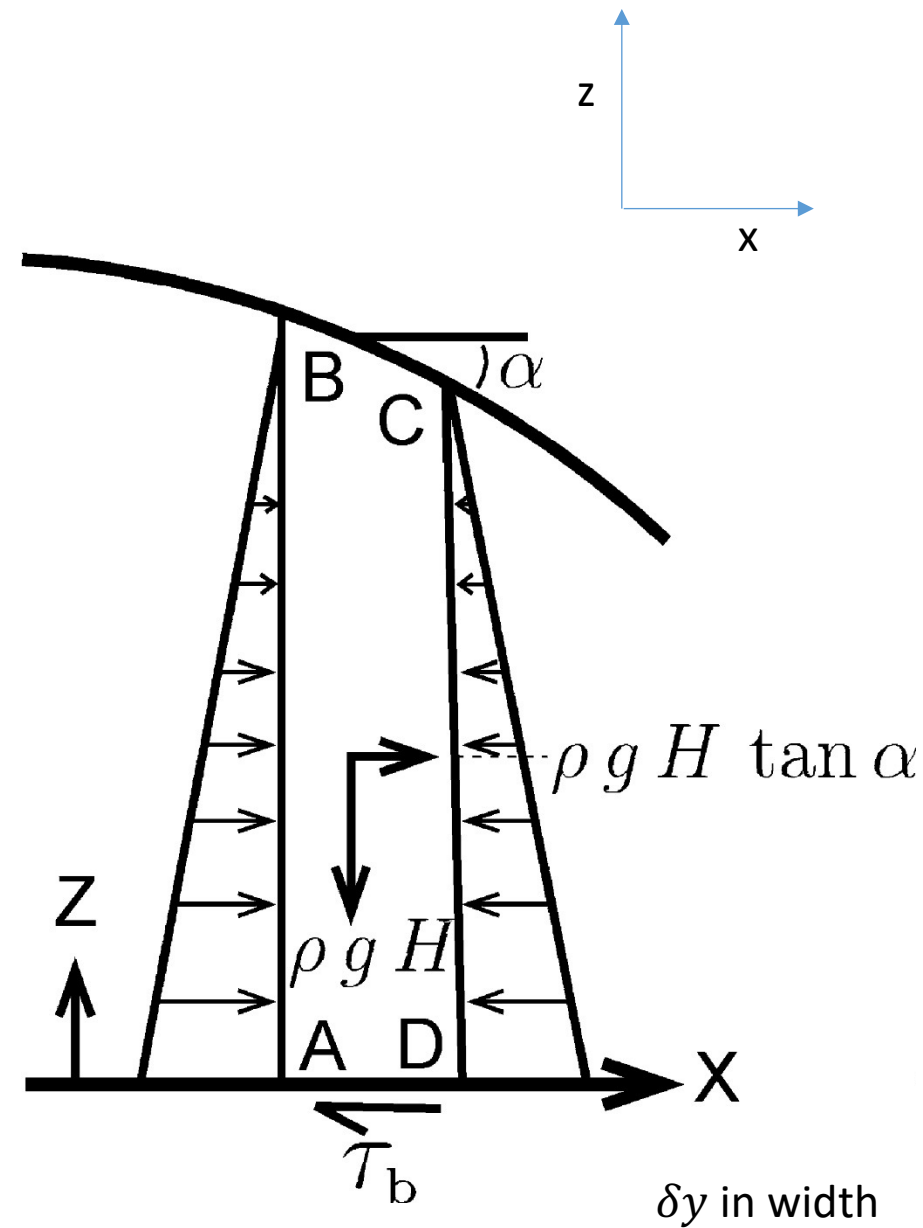
References

- DeConto, Robert M., and David Pollard.
"Contribution of Antarctica to past and future sea-level rise." *Nature* 531, no. 7596 (2016): 591.
- Hindmarsh, R.C.A., 2004. A numerical comparison of approximations to the Stokes equations used in ice sheet and glacier modeling. *Journal of Geophysical Research: Earth Surface*, 109(F1).

An alternative, simpler derivation

Driving stress

$$\tau_d = -\rho g H \frac{\partial H}{\partial x}$$

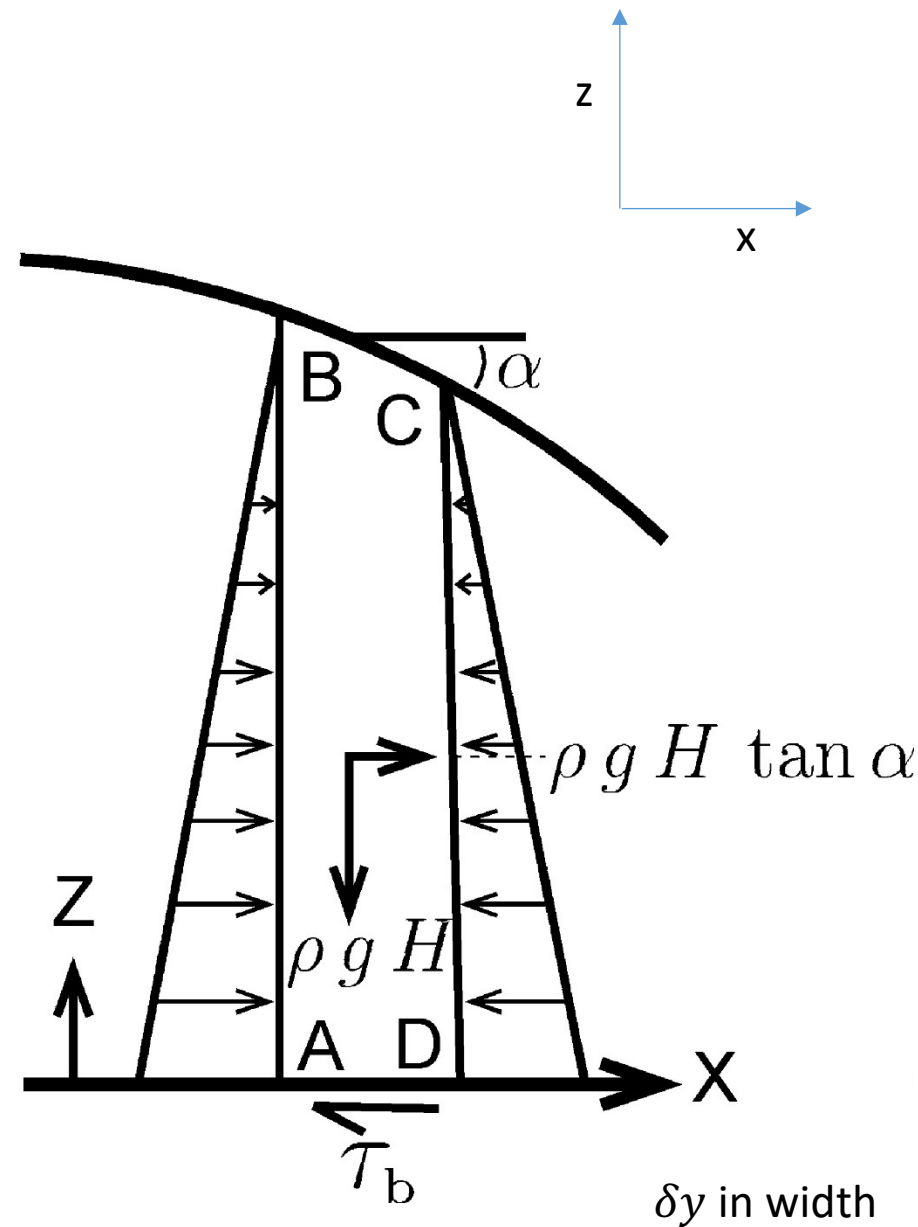


Driving stress

$$\tau_d = -\rho g H \frac{\partial H}{\partial x}$$

Total horizontal force acting in one direction integrated over the ice thickness

$$\int_0^H \delta y \rho g (H - z) dz = \frac{1}{2} \delta y \rho g H^2$$



Driving stress

$$\tau_d = -\rho g H \frac{\partial H}{\partial x}$$

Total horizontal force acting in one direction integrated over the ice thickness

$$\int_0^H \delta y \rho g (H - z) dz = \frac{1}{2} \delta y \rho g H^2$$

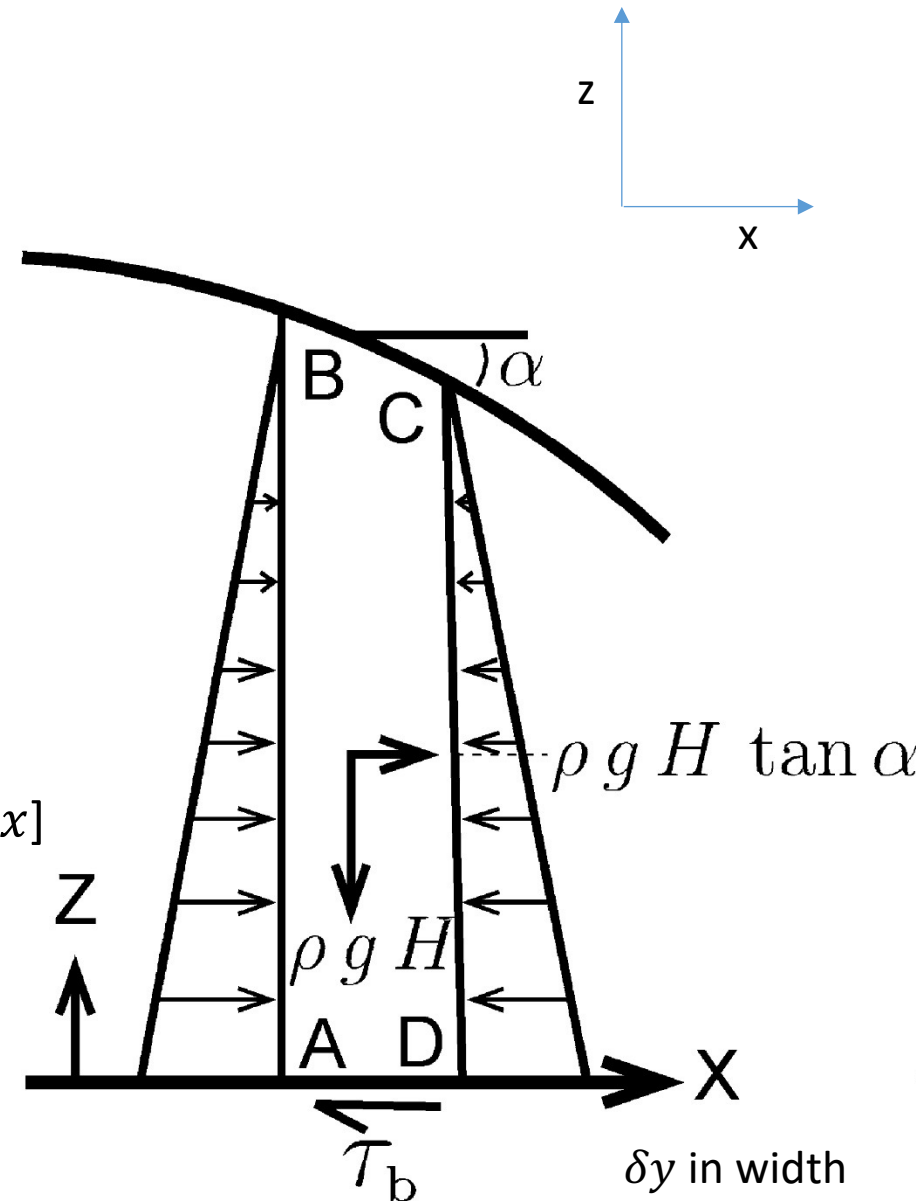
net horizontal force

$$\frac{1}{2} \delta y \rho g H^2 - \left[\frac{1}{2} \delta y \rho g H^2 + \frac{\partial}{\partial x} \left(\frac{1}{2} \delta y \rho g H^2 \right) \delta x \right]$$

$$= -\frac{\partial}{\partial x} \left(\frac{1}{2} \delta y \rho g H^2 \right) \delta x$$

Driving force per unit area = driving stress

$$\tau_d = -\rho g H \frac{\partial H}{\partial x}$$



Driving stress

$$\tau_d = -\rho g H \frac{\partial H}{\partial x} \longrightarrow \tau_d = \rho g H \alpha$$

$$\alpha = -\frac{\partial H}{\partial x}$$

Viscous stresses

We assume the driving stress is balanced by the viscous stresses in the ice.

And we assume that the driving stress is distributed linearly with depth.

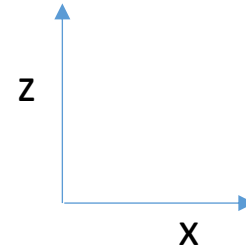
driving stress

$$\tau_d = \rho g \alpha H$$

viscous stresses in the ice

$$\tau_{zx}(z) = \rho g \alpha H \left(1 - \frac{z}{H}\right)$$

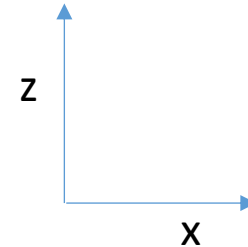
Proof of this: https://ldeo-glaciology.github.io/glaciology-intro-book/sections/ice_flow/sia_derivation.html#integrate-vertically



Rheology

Next, we use Glen's flow law to relate these stresses to the shear strain rate ϵ_{zx} .

$$\tau_{zx}(z) = \rho g \alpha H \left(1 - \frac{z}{H}\right)$$



Glen's flow law

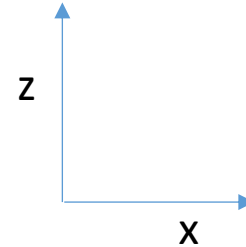
$$\epsilon_{zx}^{\dot{}} = A \tau_{zx}^n$$

$$\epsilon_{zx}^{\dot{}} = A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n$$

Strain rate $\rightarrow$ velocity gradient

Now we use the definition of strain rate to relate this to the vertical gradient in horizontal velocity.

$$\epsilon_{zx}^{\dot{}} = A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n$$



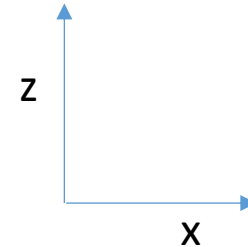
Strain rate

$$\epsilon_{zx}^{\dot{}} = \frac{1}{2} \left[\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right]$$

Strain rate $\rightarrow$ velocity gradient

Now we use the definition of strain rate to relate this to the vertical gradient in horizontal velocity.

$$\epsilon_{zx}^{\dot{}} = A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n$$



Strain rate

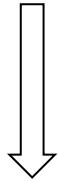
$$\epsilon_{zx}^{\dot{}} = \frac{1}{2} \left[\cancel{\frac{\partial w}{\partial x}} + \frac{\partial u}{\partial z} \right]$$

$$\epsilon_{zx}^{\dot{}} = \frac{1}{2} \frac{\partial u}{\partial z}$$

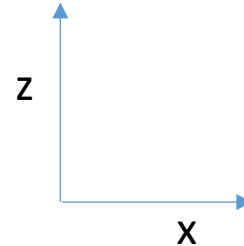
Strain rate $\rightarrow$ velocity gradient

Now we use the definition of strain rate to relate this to the vertical gradient in horizontal velocity.

$$\epsilon_{zx}^{\dot{}} = A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n$$



$$\frac{\partial u}{\partial z} = 2A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n$$



Strain rate

$$\epsilon_{zx}^{\dot{}} = \frac{1}{2} \left[\cancel{\frac{\partial w}{\partial x}} + \frac{\partial u}{\partial z} \right]$$

$$\epsilon_{zx}^{\dot{}} = \frac{1}{2} \frac{\partial u}{\partial z}$$

Horizontal velocity

We next integrate vertically to get the horizontal velocity as a function of depth.

$$\frac{\partial u}{\partial z} = 2A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n \quad \zeta = 1 - \frac{z}{H}$$

Integrate by substitution

$$u(z) = 2A(\rho g \alpha H)^n \left(\frac{H}{n+1}\right) \left(1 - \left(1 - \frac{z}{H}\right)^{n+1}\right)$$

Ice flux

We next integrate vertically again to get ice flux q

$$\frac{\partial u}{\partial z} = 2A(\rho g \alpha H)^n \left(1 - \frac{z}{H}\right)^n \quad \zeta = 1 - \frac{z}{H}$$

Integrate by substitution

$$u(z) = 2A(\rho g \alpha H)^n \left(\frac{H}{n+1}\right) \left(1 - \left(1 - \frac{z}{H}\right)^{n+1}\right)$$

Integrate one more time to get depth-integrated volume-flux per unit width:

$$q = \frac{2A}{n+2} (\rho g \alpha)^n H^{n+2}$$

Full derivation here: https://ideo-glaciology.github.io/glaciology-intro-book/sections/ice_flow/sia_derivation.html